

Automatic Light Detection System Using LDR and Arduino

Objective

The objective of this project is to simulate an IoT-based light detection system using an LDR (Light Dependent Resistor) and Arduino Uno. The system detects changes in ambient light intensity and provides sensor readings that can be used to control devices such as automatic lights in smart homes and street lighting systems.

Components Used

The following components were used in this simulation:

Arduino Uno R3, Light Dependent Resistor (LDR) ,10k Ω Resistor ,LED ,220 Ω Resistor ,Breadboard Jumper Wires, Tinkercad Simulator

Circuit Description

The circuit was designed using Tinkercad. The LDR and a 10k Ω resistor were connected as a voltage divider circuit. The output of the voltage divider was connected to the analog input pin A0 of the Arduino Uno. The Arduino continuously reads the sensor value and displays it through the Serial Monitor.

An LED was also included in the circuit to represent a lighting device that can be controlled based on the detected light level. The Arduino processes the sensor data and can be programmed to switch the LED on or off depending on the ambient light intensity.

Arduino Code

```
int ldr = A0;

int led = 8;

void setup()
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int value = analogRead(ldr);
  Serial.println(value);
  if (value < 500) { // Dark
    digitalWrite(led, HIGH);
  } else {          // Bright
    digitalWrite(led, LOW);
  }
  delay(100);
}
```

Arduino Code Explanation

The program begins by declaring two variables. The variable `ldr` is assigned to analog pin A0, which is used to read data from the Light Dependent Resistor (LDR). The variable `led` is assigned to digital pin 8, which is connected to the LED.

In the `setup()` function, the LED pin is configured as an output using `pinMode(led, OUTPUT)`. The Serial Monitor is also initialized with a baud rate of 9600 using `Serial.begin(9600)`, allowing sensor values to be displayed for monitoring purposes.

Inside the `loop()` function, the Arduino continuously reads the analog value from the LDR using `analogRead(ldr)` and stores it in the variable `value`. This value is then printed to the Serial Monitor using `Serial.println(value)`.

The program uses an if-else statement to determine whether the environment is dark or bright. If the sensor value is less than 500, the environment is considered dark, and the LED is turned ON using `digitalWrite(led, HIGH)`. If the sensor value is greater than or equal to 500, the environment is considered bright, and the LED is turned OFF using `digitalWrite(led, LOW)`.

Finally, a delay of 100 milliseconds is added using `delay(100)` to ensure stable readings and prevent excessive processing.

This code demonstrates how an LDR sensor can be used to automatically control lighting based on ambient light intensity, similar to smart street lights and smart home lighting systems.

Working Principle

The LDR is a light-sensitive resistor whose resistance changes according to the intensity of light falling on its surface. When the environment is bright, the resistance of the LDR decreases, resulting in higher analog readings. When the environment becomes dark, the resistance increases, causing lower analog readings.

The Arduino reads these values through analog pin A0 and displays them on the Serial Monitor. During the simulation, the sensor readings changed significantly when the light intensity was adjusted, demonstrating the successful operation of the light detection system.

Simulation Results

The simulation produced the following observations:

- Bright Condition: Sensor reading approximately 932
- Dark Condition: Sensor reading approximately 54
- The Arduino successfully detected changes in light intensity.
- The Serial Monitor displayed real-time sensor values.

These results confirm that the LDR sensor accurately responds to variations in ambient light levels.

Applications

This type of IoT-based light detection system can be used in:

- Automatic street lighting systems
- Smart home lighting automation
- Energy-saving lighting control systems
- Industrial monitoring systems
- Smart city infrastructure

Conclusion

This project successfully demonstrated a sensor-based IoT simulation using an LDR and Arduino Uno in Tinkercad. The LDR accurately detected changes in ambient light intensity, and the Arduino successfully processed the sensor data. The simulation highlights the practical use of light sensors in smart automation systems and demonstrates a simple yet effective IoT application.

Screenshots

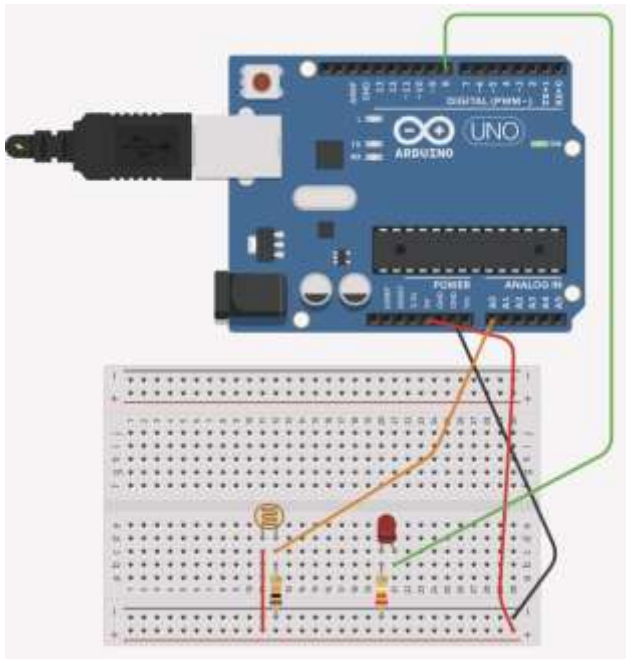


Figure 2: Complete Circuit Diagram

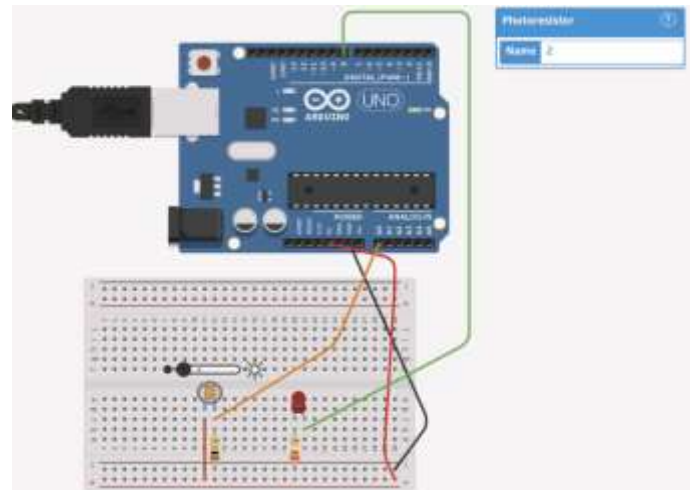


Figure 1: Bright Condition (Sensor Reading ≈ 932)

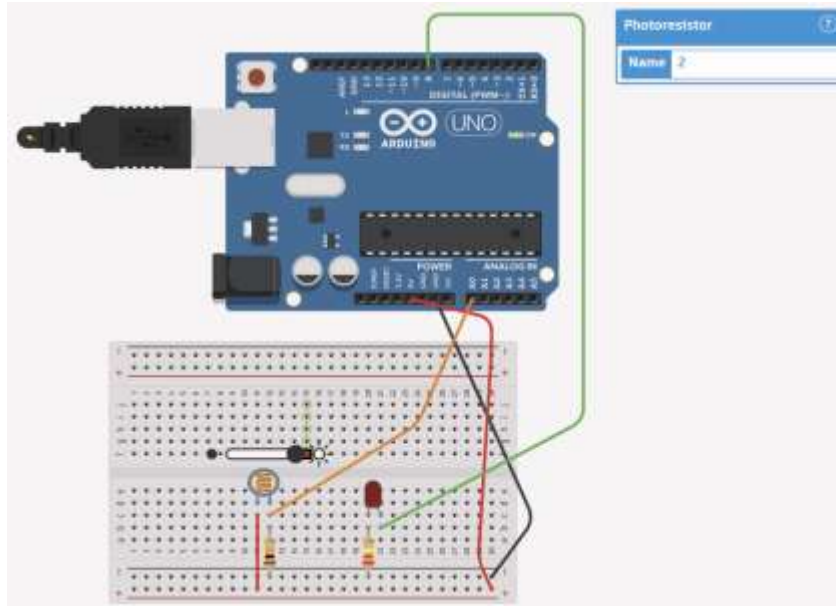


Figure 3: Dark Condition (Sensor Reading ≈ 54)

```

1  int ldr = A0;
2  int led = 8;
3
4  void setup() {
5      pinMode(led, OUTPUT);
6      Serial.begin(9600);
7  }
8
9  void loop() {
10     int value = analogRead(ldr);
11
12     Serial.println(value);
13
14     if (value < 500) {    // Dark
15         digitalWrite(led, HIGH);
16     } else {             // Bright
17         digitalWrite(led, LOW);
18     }
19
20     delay(100);
21 }

```

 Serial Monitor

951
951
951
951

Figure 4: Arduino Code and Serial Monitor Output